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Frequency-Dependent Fault Identification by Convolutional Neural Networks with Uncertainty Analysis

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Abstract

Frequency-dependent similarity has previously been used to preferentially detect faults with different vertical displacements at different dominant frequencies (e.g. Dewett and Henza, 2016; Johan and Castagna, 2017). By decomposing seismic data into multi-frequency dominated data, structural and stratigraphic delineation such as complex fault systems could be better interpreted in various scales. Similarities and differences are also highlighted. Low-frequency seismic data provides an effective broadband to suppress wavelet partials and improve the resolution of seismic data. Several authors have pointed out that low-frequency data offers a stronger penetration ability to restrict the influence of scattering and absorption on the reflected waves (e.g. Wu, et al., 2013; Sun et al., 2021). On the other hand, high-frequency seismic data could improve vertical resolution, leading to a better mapping of horizons and stratigraphic features that are key for a detailed geological characterization of the reservoirs.

Convolutional Neural Networks (CNN) show a promising way to automatically identify fault attributes in seismic data by analyzing image segmentation and feature extraction with high accuracy. Wu et al. (2019) performed efficient image-to-image fault segmentation by using a supervised CNN architecture. By creating multiple synthetic seismic images and corresponding binary fault-labeled images, the authors successfully trained a fault segmentation network. Jiang and Norlund (2020, 2021) implemented a multi-channel CNN with a multi-scale aggregated module to combine pixel-level accuracy with global-level feature identification to improve prediction accuracy. However, those prediction images may still be disappointing due to strong noise existing in the data or the juxtaposition of relatively similar reflectors across a fault. Lyu et al. (2019) computed multiple spectral coherence to reduce the noise and improved the continuity of the faults.

Quantifying uncertainty in seismic interpretation can be divided into regression settings, such as rock property analysis, or classification setting, such as semantic segmentation. There are two main types of uncertainty to evaluate a deep learning model. Aleatoric uncertainty captures noise inherent randomness of distribution in the machine learning prediction. The prototypical example of aleatoric uncertainty is coin flipping. The data generating process in this type of experiment has a stochastic component that cannot be reduced by any additional source of information. Conversely, epistemic uncertainty stands for uncertainty caused by a lack of knowledge to obtain a best model. It refers to the ignorance of

the agent or decision maker, and hence to the epistemic state of the agent instead of any underlying random phenomenon. (Hullermeier and Waegeman, 2021). Kendall and Gal (2017) show the uncertainty observations in many different big data regimes to demonstrate the efficiency of uncertainty analysis in computer vision applications. Kwon et al (2020) implemented a Bayesian neural network and propose a natural way to quantify uncertainty in classification problems by decamping predictive uncertainty into two parts, aleatoric and epistemic uncertainty.

In this paper, we performed spectral decomposition to decompose a group of training seismic data to different frequencies. We then fed each iso-frequency training data into a multi-channel, multi-scale CNN architecture to train a series of frequency-dependent ML models to evaluate the ability to enhance the fault identification process. We applied our technique to a real case study, revealing significant improvements on the prediction results using a multiple frequencies model compared to those predictions generated from a full-bandwidth ML model

Introduction

Frequency-dependent similarity has previously been used to preferentially detect faults with different vertical displacements at different dominant frequencies (e.g. Dewett and Henza, 2016; Johan and Castagna, 2017). By decomposing seismic data into multi-frequency dominated data, structural and stratigraphic delineation such as complex fault systems could be better interpreted in various scales Figure (1). Figure (2) described the frequency spectrum of the original seismic data and frequency spectrum of the decomposed seismic data. Similarities and differences are also highlighted. Low-frequency seismic data provides an effective broadband to suppress wavelet partials and improve the resolution of seismic data. Several authors have pointed out that low-frequency data offers a stronger penetration ability to restrict the influence of scattering and absorption on the reflected waves (e.g. Wu, et al., 2013; Sun et al., 2021). On the other hand, high-frequency seismic data could improve vertical resolution, leading to a better mapping of horizons and stratigraphic features that are key for a detailed geological characterization of the reservoirs.

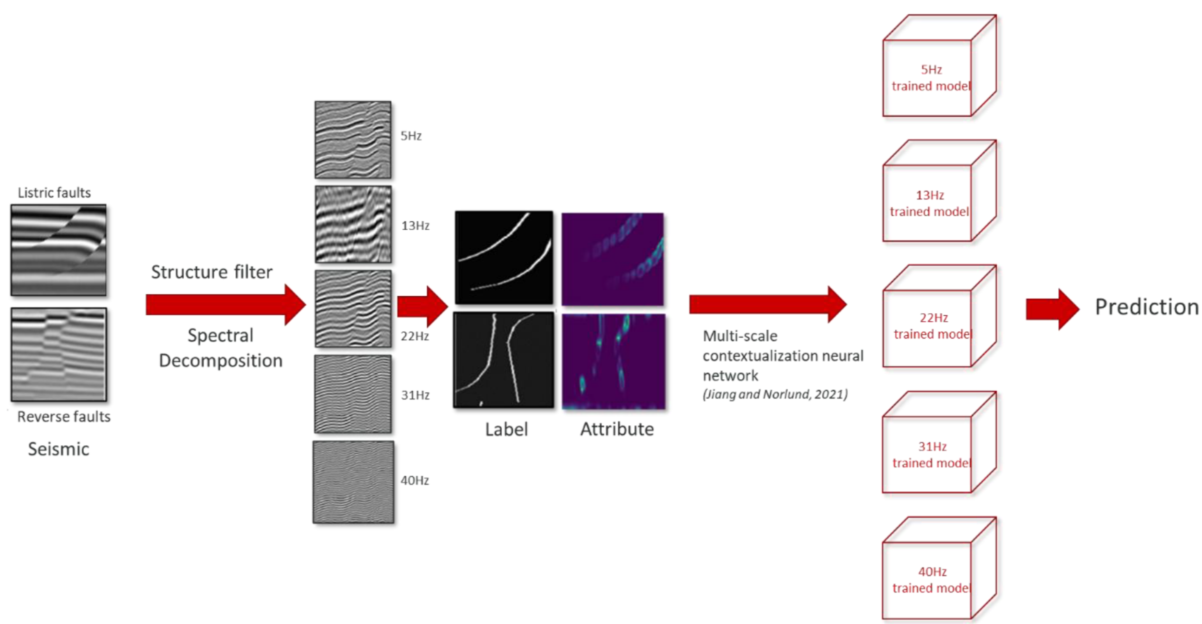

Figure 1—The workflow of Frequency-dependent Fault Interpretation

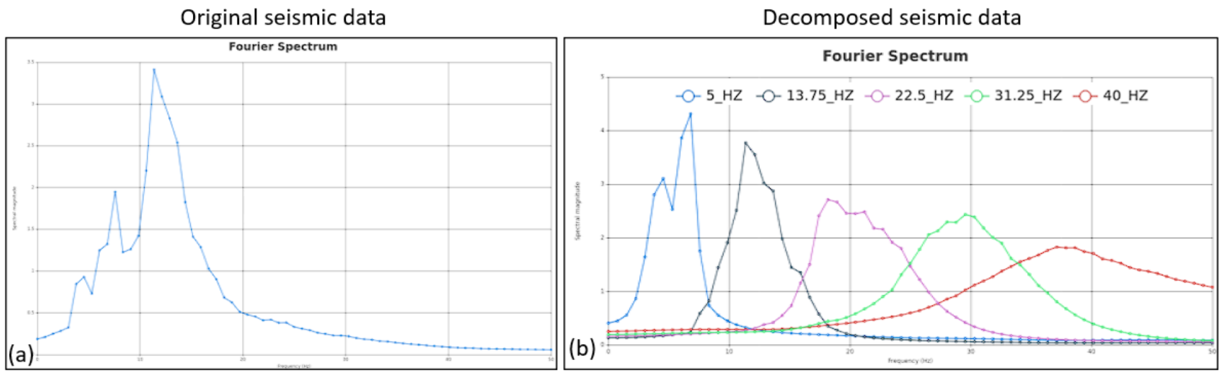

Figure 2—(a) Frequency spectrum of the original training data, (b) frequency spectrum of the original training data

Convolutional Neural Networks (CNN) show a promising way to automatically identify fault attributes in seismic data by analysing image segmentation and feature extraction with high accuracy. Wu et al. (2019) performed an efficient image-to-image fault segmentation by using a supervised CNN architecture. By creating multiple synthetic seismic images and corresponding binary fault-labeled images, the authors successfully trained a fault segmentation network. Jiang and Norlund (2020, 2021) implemented a multi-channel CNN with a multi-scale aggregated module to combine pixel-level accuracy with global-level feature identification to improve the prediction accuracy. This dilated convolution has been shown the ability to detect finer details by bringing a broader view of the input to capture more neighbouring information. In 1D, dilated convolution is defined as:

$$output[i] = \sum_{l=1}^{L} input[i + r * l] * h[l] \tag{1}$$

Where *input* and *output* are extracted seismic features in CNN. $h[l]$ represents the convolution filter of length L, and r denotes the dilation rate we use to sample $input[i]$. A standard convolution will have the dilation rate equals 1. We then combine standard convolution and dilated convolution in 3D CNN, followed by a multi-scale aggregated module, to calculate a full range of seismic receptive fields. Their multi-scale module is based on a rectangular prism of a set of dilated convolutional layers, which support the exponential expansion of the receptive field without losing resolution or coverage. This module takes the feature maps produced from the previous layer as input and outputs the same form of the feature maps. Since the input and output have the same form, it is convenient to plug the multi-scale module into other existing fault prediction networks.

However, those prediction images may still be disappointing, falling short of providing a clear and geologically plausible fault network. This deficiency often stems from two primary challenges inherent in seismic data. First, the presence of strong random or coherent noise, such as acquisition footprints or migration artifacts, can obscure the subtle waveform changes associated with faulting, leading to a "blotchy" or ambiguous attribute response. Second, and perhaps more fundamentally, is the geological challenge of the juxtaposition of relatively similar reflectors across a fault. Classic coherence and similarity algorithms are designed to detect lateral discontinuities in the seismic waveform (Marfurt et al., 1998), but they can fail when the acoustic impedance contrast across the fault is low, rendering the fault plane nearly transparent to the algorithm. To specifically address these issues, Lyu et al. (2019) computed multiple spectral coherence to reduce the noise and improved the continuity of the faults. Rather than calculating coherence over the entire seismic spectrum where noise might dominate, their approach leverages spectral decomposition. By computing coherence on multiple, narrow frequency bands and then combining the results, they effectively enhance the signal-to-noise ratio, amplifying the true fault signature that may only be prominent at specific frequencies (Chopra and Marfurt, 2007). The outcome of this multi-spectral approach is a significant reduction in noise artifacts and, crucially, improved continuity and clarity of the fault images, even in areas of low reflector contrast, thereby making subsequent structural interpretation more reliable.

Quantifying uncertainty in seismic interpretation can be divided into regression setting, such as rock property analysis, or classification setting, such as semantic segmentation. There are two main types of uncertainty to evaluate a deep learning model. Aleatoric uncertainty captures noise inherent randomness of distribution in the machine learning prediction. The prototypical example of aleatoric uncertainty is coin flipping. The data generating process in this type of experiment has a stochastic component that cannot be reduced by any additional source of information. Conversely, epistemic uncertainty stands for uncertainty caused by a lack of knowledge to obtain a best model. It refers to the ignorance of the agent or decision maker, and hence to the epistemic state of the agent instead of any underlying random phenomenon. (Hullermeier and Waegeman, 2021). Kendall and Gal (2017) show the uncertainty observations in many different big data regimes to demonstrate the efficiency of uncertainty analysis in computer vision application. Kwon et al (2020) implemented a Bayesian neural network and propose a natural way to quantify uncertainty in classification problems by decamping predictive uncertainty into two parts, aleatoric and epistemic uncertainty.

In this paper, we performed spectral decomposition to decompose a group of training seismic data to different frequencies. We then fed each iso-frequency training data into a multi-channel, multi-scale CNN architecture to train a series of frequency-dependent ML models to evaluate the ability to enhance the fault identification process. We applied our technique to a real case study, revealing significant improvements on the prediction results using a multiple frequencies model compared to those predictions generated from a full-bandwidth ML model

Workflow & Methodology

Extracting geologic features from seismic data is crucial but time-consuming work. Researchers have developed different deterministic algorithms to analyze the seismic information, such as amplitude variation with offset (AVO) for amplitude analysis, source mechanism for phase analysis, or spectral decomposition for frequency analysis. In this paper, we applied a Morlet wavelet over input synthetic traces to decompose each trace by frequencies. We chose four main frequencies to decompose a group of input synthetic data. Figure 2 shows frequency spectrum of Figure (1) from full-band and decomposed seismic data. The full band data has a central frequency of 10Hz. We decomposed this into four different frequencies: 5Hz, 13Hz, 22Hz, and 31Hz. Notice that low-frequency data, such as 5Hz, shows fewer details (Figure 1) but more general trends with longer-wavelength. This low-frequency training data could potentially help us explore large-scale faults, or the first order faults, but ignore small-scale fracture-like faults.

A central challenge in applying advanced machine learning models like FDFI to real-world datasets is the sheer scale of modern seismic volumes, which can often reach terabytes in size. To address this, we strategically moved beyond single-machine processing and leveraged the high-performance computing (HPC) capabilities at ADNOC through a data parallelism framework. This approach is exceptionally well-suited for seismic inference tasks, which are often "embarrassingly parallel" in nature. The core principle involves partitioning the large-scale seismic volume into numerous smaller, independent sub-volumes or batches of seismic traces (as depicted in Figure 3). Each of these data partitions is then distributed to a different compute node within the HPC cluster. Crucially, each node—whether CPU or GPU-based—loads an identical, independent copy of the pretrained FDFI model and performs inference only on the data chunk it receives. Because there are no dependencies between the sub-volumes, all nodes can process their assigned data simultaneously. Once inference is complete on all partitions, the resulting fault predictions are reassembled into a single, cohesive volume, preserving the correct spatial relationships. This architecture perfectly aligns with the design of modern HPC clusters, transforming a monolithic, time-consuming task into a highly scalable, parallelized workflow.

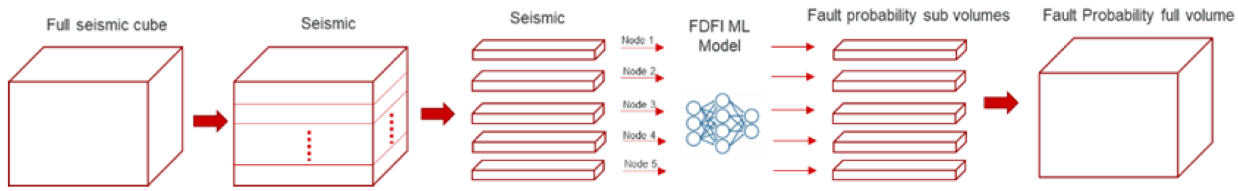

Figure 3—The FDFI architecture of parallel computing on HPC capability

The performance impact of this data-parallel strategy was transformative, yielding an acceleration factor of 20 when compared to traditional multithreading on a single, powerful node. This dramatic improvement stems from our ability to scale horizontally, thereby overcoming the inherent limitations of a single machine, such as finite memory, I/O bandwidth, and core count. For ADNOC, this efficiency gain is not merely an incremental improvement; it represents a paradigm shift in the utility of ML-driven interpretation. It reduces the turnaround time for generating a full-volume fault analysis from potentially weeks to mere hours or days. This acceleration enables a more dynamic and iterative interpretation workflow, where geoscientists can rapidly test hypotheses, re-run the model with different parameters or frequency inputs, and integrate the results into their projects without prohibitive delays. Ultimately, leveraging data parallelism on HPC infrastructure transforms the FDFI model from a specialized, computationally expensive tool into a highly operational and scalable solution, providing a significant competitive advantage in accelerating the exploration and development cycle.

A key advantage of the FDFI methodology is its inherent frequency-dependent nature, which allows the interpreter to strategically target faults at a desired geological scale. While seismic data contains a broad spectrum of frequencies, these different frequencies illuminate different aspects of the subsurface. High frequencies are essential for resolving fine details, such as thin beds and minor fractures, but they are also more susceptible to noise and can produce an overwhelmingly complex fault image cluttered with small-scale fault branches and subsidiary splays. Conversely, low frequencies attenuate fine details but excel at imaging large-scale structural architecture, revealing the continuity and connectivity of the primary fault network. Figure 4 shows several channels were delineated by vertical faults. This is also an interesting part that FDFI was designed to extract faults from seismic data but not channel itself. However, the natural of channel was composed of steep-dip boundary to form a fluvial channel. FDFI in this case also picked the boundary as small-scale faults but not the entire channel body, especially the bottom of channel. It is still required to use any channel segmentation algorithm to extract channel bodies from seismic data. Our objective is to delineate the major, field-defining faults and avoid the interpretive noise of small, insignificant branches. As a result, the FDFI output effectively highlights the core fault system while suppressing the high-frequency clutter, leading to a cleaner, more robust, and geologically meaningful structural interpretation that directly aligns with our large-scale objectives. This workflow transforms FDFI from a simple fault detector into a tunable interpretation tool, where the choice of input frequency band becomes a direct proxy for controlling the scale of the investigation (Figure 4) .

The ability of FDFI is to have control on the scale of faults. Since the models were trained by different frequency-dependent data, we are capable of image the large-scale fault which contains low-frequency or the small-scale fault which contains high frequency. By decomposing seismic data into multi-frequency dominated data, structural and stratigraphic delineation such as complex fault systems could be better interpreted in various scales. Similarities and differences are also highlighted. Low-frequency seismic data provides an effective broadband to suppress wavelet partials and improve the resolution of seismic data. We perform FDFI on the same ADNOC data but with different pre-trained models. The low-frequency models will skip imagining those low-scale faults as it is only sensitive to the large-scale fault, or the first-order fault. It could be useful when we want to trim small-scale fault branch while clearly image the main fault segment.

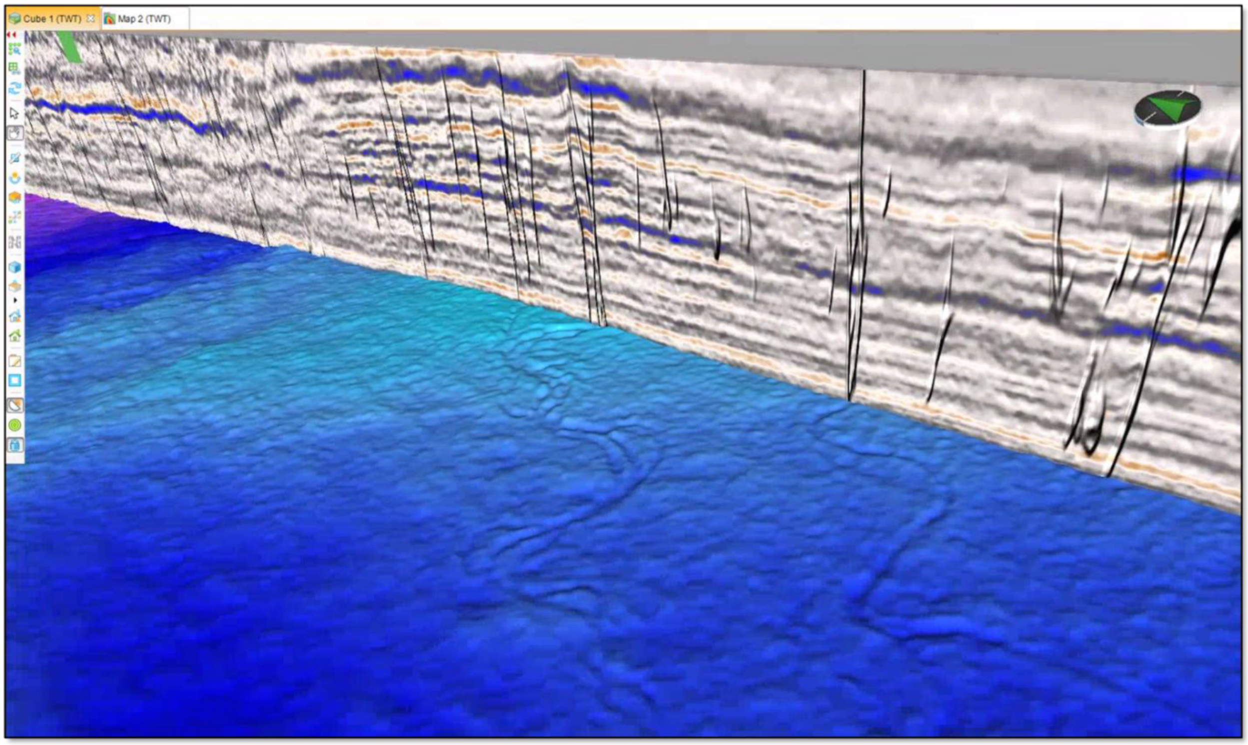

Figure 4—A cross-section of fault overlay on seismic with RUS horizon

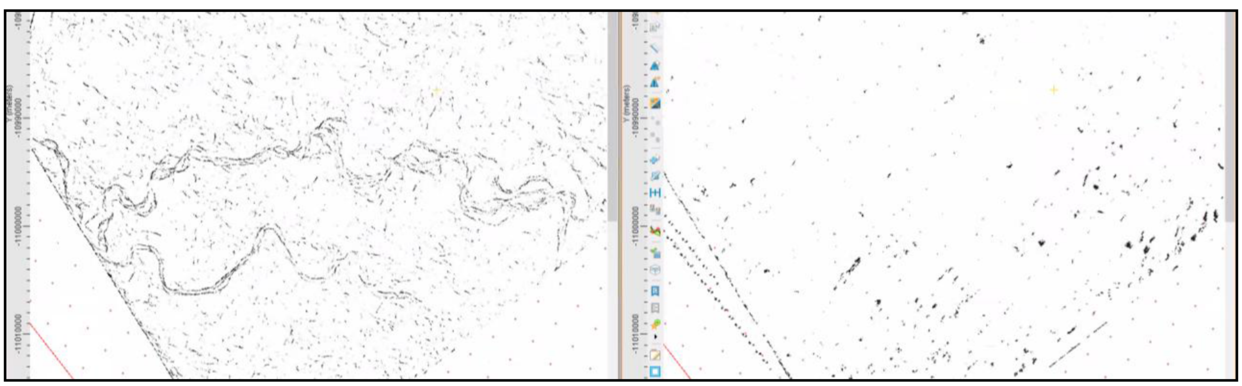

Figure 5—Fault prediction comparison between the full-frequency ML model and the low-frequency ML model. Low-frequency ML model could help to control the scale of expected faults. (Left) the small-scale faults image channels on the map view by a full-frequency ml MODEL; (Right) the low-frequency ML model will skip to image the small-scale faults.

During the work, we have found at the shallow depth, the faults show a clear NE-SW trend (Figure 6). From Noufal et al., (2016) the secondary contractional and extensional structures typically form at predictable angles to the orientation of the strike-slip zone. In map view the maximum and minimum compressional stress are oriented at approximately 45 degrees to the strike-slip zone and can therefore be used as kinematic indicators for determining the sense of movement on the fault zone. Compared with Abu Dhabi Tectonic map, the fault trend direction is aligned with stress regime, this is important to determine the relative age of the structures relative to the strike-slip fault zone.

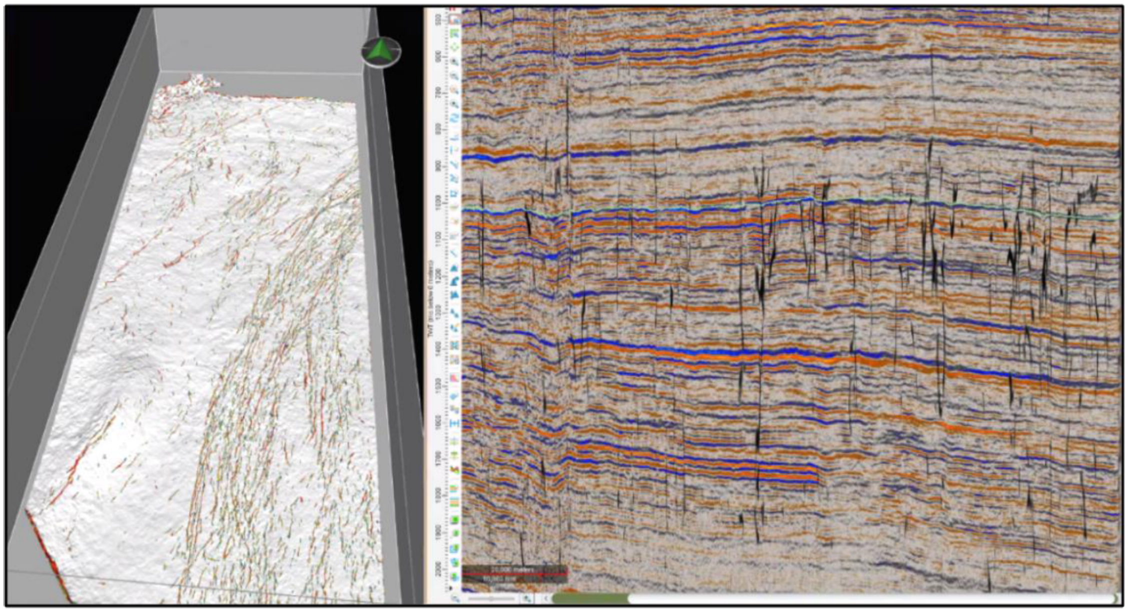

Figure 6—(Left) FDFI faults overlay on horizon FIQA; (Right) The cross-section of faults overlay on seismic data

Figure 6 shows an overlay of the extracted faults on the deeper horizon. On Ghayati mega survey, we observed a clearly 4-way dip faulting system (Figure 7(a)). Four-way dip closures refer to geological structures where rock layers are folded in such a way that they dip toward a central point from four different directions. This often forms a closure or a trap for hydrocarbons, making it an important feature in petroleum geology and exploration. In these structures, the layers of rock can create an effective seal, preventing oil or gas from migrating out. The four-way closure is significant because it provides a more robust trap compared to two-way closures, which only have a dip in two directions. Geologists often look for these formations when assessing potential oil and gas reservoirs. Most of the discovered large onshore fields to date in this area are high-relief, four-way dip closures over basement highs-oriented in NE-SW direction. However, within the area, the expected trap geometries include also fault segmented low-relief four-way dip closures and three-way fault-dependent dip closures. In addition to the small-size structure traps there is a potential of stratigraphic trapping within the Clinoforms of Shuaiba and Habshan formations. A high-quality 3D faulting system with improved vertical and lateral resolution is crucial to unravel those subtle traps and other interpretation tasks.

Figure 7(b) also shows a multi-way dip faulting system on Onshore mega survey. The master fault system shows differential uplifting and subsidence along the strike relative to the transpressional and transtensional regimes that act between the fault segments. The maximum amount of the uplift is recorded along the master fault segment where it forms four-way dip closures. We may consider perform multi-azimuthal FDFI to bring helps for fracture and faults characterization on the final volume.

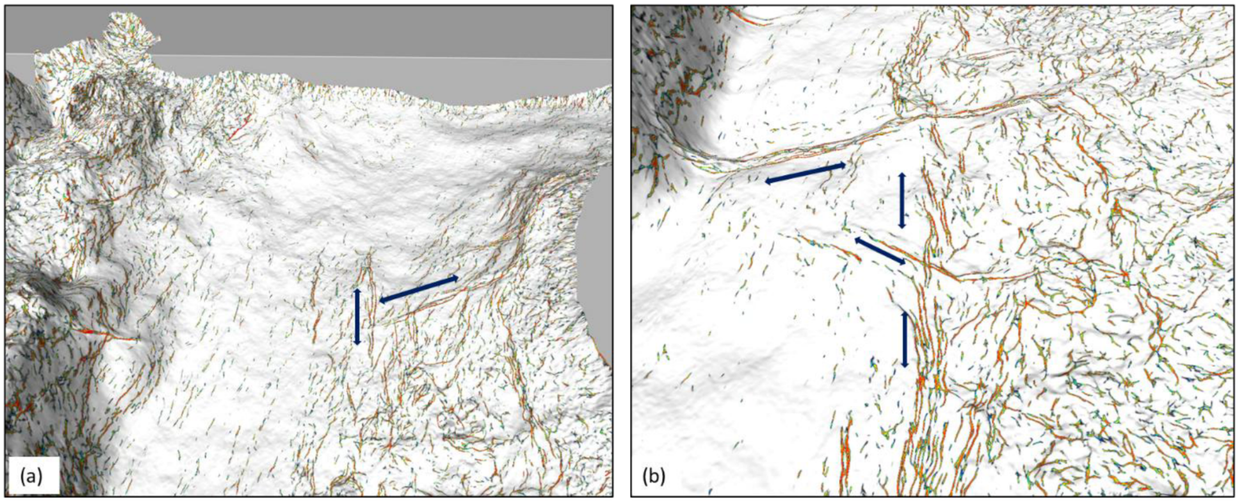

Figure 7—(Top) An overview of the new extracted horizon below Unazyah; (Middle) FDFI faults overlay on the new horizon at Onshore Block 1 survey; (Bottom) FDFI faults on Onshore Block 2 survey.

Discussions

The use of frequency-dependent fault interpretation tools offers a major benefit in addressing the structural problems in the United Arab Emirates, especially in the basins of Abu Dhabi that are dominated by carbonate and evaporite. The region's geology is distinguished by low-displacement, incredibly subtle faults that are crucial for reservoir compartmentalization but frequently obscured on full-bandwidth seismic data. The ability to improve the signal-to-noise ratio for these particular targets is the main advantage of a tunable, frequency-based machine learning tool. By examining a low-frequency volume (e.g. A. interpreters are better able to penetrate through complex overburden and define the large-scale fault architecture linked to salt tectonics or deep-seated structures (15-30 Hz). Higher-frequency bands, on the other hand, can be used to isolate and enhance the expression of small intra-reservoir faults, which are crucial for comprehending fluid flow.

But the effectiveness of these sophisticated tools is not automatic; they introduce important factors that require geological knowledge and methodical workflow. Model bias is the biggest risk because a large number of pre-trained models are created using clastic datasets (e.g. A. Gulf of Mexico), and could result in geologically irrational artifacts by misinterpreting the seismic response of carbonate facies as well. As a result, thorough verification and possible adjustment on local data are necessary. Additionally, the tool's tunability may result in interpreter bias, where a frequency band may be unconsciously selected to support an existing theory rather than to present the most objective outcome. A methodical examination across several frequency bands is required for this. Lastly, while low frequencies are great for producing a clear, regional map, they run the risk of oversimplifying intricate fracture systems by leaving out smaller-scale faults that may be crucial for reservoir permeability

Results

We ran FDFI through ADNOC's merged seismic data to predict fault probability map. The prevailing regime in Abu Dhabi is the strike-slip, as the older faults are those parallel to the Najd fault system. Most of the faults in the seismic data of Onshore Block 2 and Gayathi show no large vertical displacement of strata or rock units on either side. The movement is dominantly horizontal, resulting in the lateral displacement of the units. We also found tear faults are also encountered in the study area and can be defined as relatively small-scale, local strike-slip faults that are associated with other structures such as folds, thrust faults and normal faults. The strike-slip fault system generated by FDFI is also verified by tectonic map of Abu Dhabi. This strike-slip system introduces a degree of structural complexity where we found some transpressional and transtentional features along the length of the fault system. From tectonic perspective, FDFI fault mapping matches well with Abu Dhabi tectonic chart published by Noufal et al., (2016):

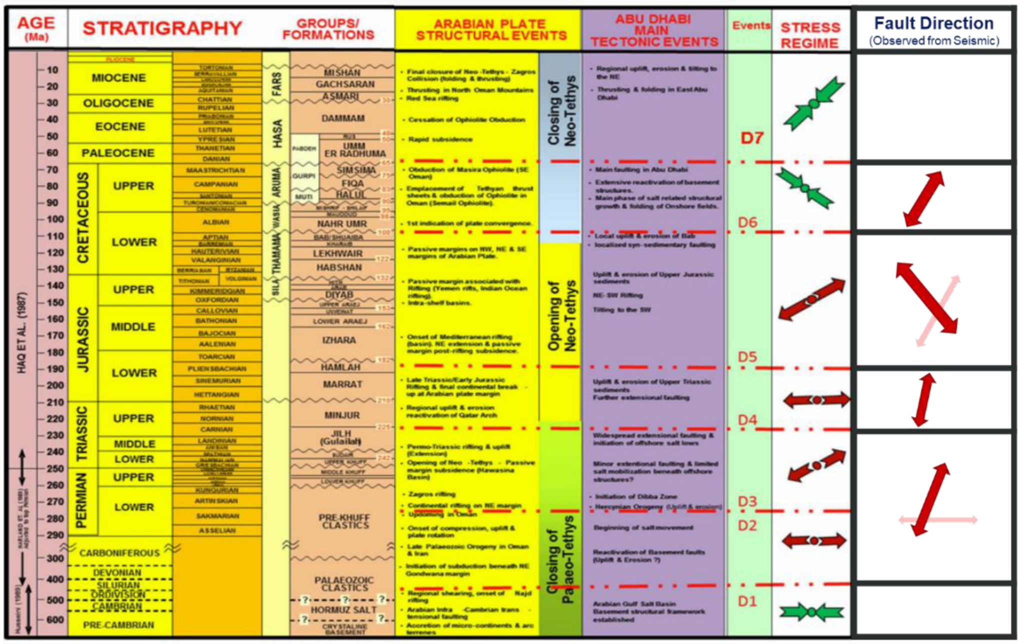

Figure 8—The tectonic map of Abu Dhabi with modification on fault direction observed by FDFI from seismic data. Image was modified from Noufal et al., 2016.

With FDFI, we could leverage the machine learning model to predict and separate channels from faults events. The predicted faults match well with the tectonic map of Abu Dhabi. Multiple multi-way dip faulting features are discovered at very deep time slice (>5.0s). the output of FDFI faults could also be used as input to AHI tool to improve horizon extraction.

The strike-slip fault system generated by FDFI is also verified by tectonic map of Abu Dhabi. This strike-slip system introduces a degree of structural complexity where we found some transpressional and transtentional features along the length of the fault system. From a tectonic perspective, FDFI fault mapping matches well with Abu Dhabi tectonic chart published by Noufal et al., (2016): The fault probability volume generated by FDFI shows that most faults are perpendicular to stress regime, which also demonstrates the fidelity of FDFI on unseen geological environment with minimal generalization gap.

Conclusions

We have presented a frequency-dependent fault prediction workflow to delineate faults from seismic data. The uncertainty analysis obtained by the predictive variance provides additional help for the machine learning-based fault prediction model. Implementation of this uncertainty as a mask significantly improved the accuracy of fault prediction and generated more continuous fault planes. A similar analysis could also be implemented in other seismic interpretation workflows, such as salt body classification and seismic facies interpretation. In the current workflow, the proposed method helps to generate large-scale, major faults in complex areas and avoids picking up small-scale fault branches which could bring extra ambiguity to geophysical interpretation. Further research will focus on combing multi-frequency predictions to add small-scale faults where we have confidence to build up a multi-scale fault probability map.

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